

Big Iron is what industry insiders call a supercomputer. We lesser mortals perceive them as massive number crunching monsters, housed in huge metallic cabinets and located in eerie blue-tinted rooms, in tightly controlled environments, spitting out bits by the billion—actually, supercomputers are mostly just that! Supercomputing can be more accurately defined as High Performance Computing (HPC). HPC covers a broad plane of various architectures and tools, used in solving large, complex and computation-intensive problems.

A lot of supercomputing's earlier history can be traced back to the late Seymour R. Cray, founder of what is now called Cray Inc., the foremost supercomputing designers of our times. He is known as the 'father of supercomputing', having founded Cray Research in 1972, after working with the CDC (Control

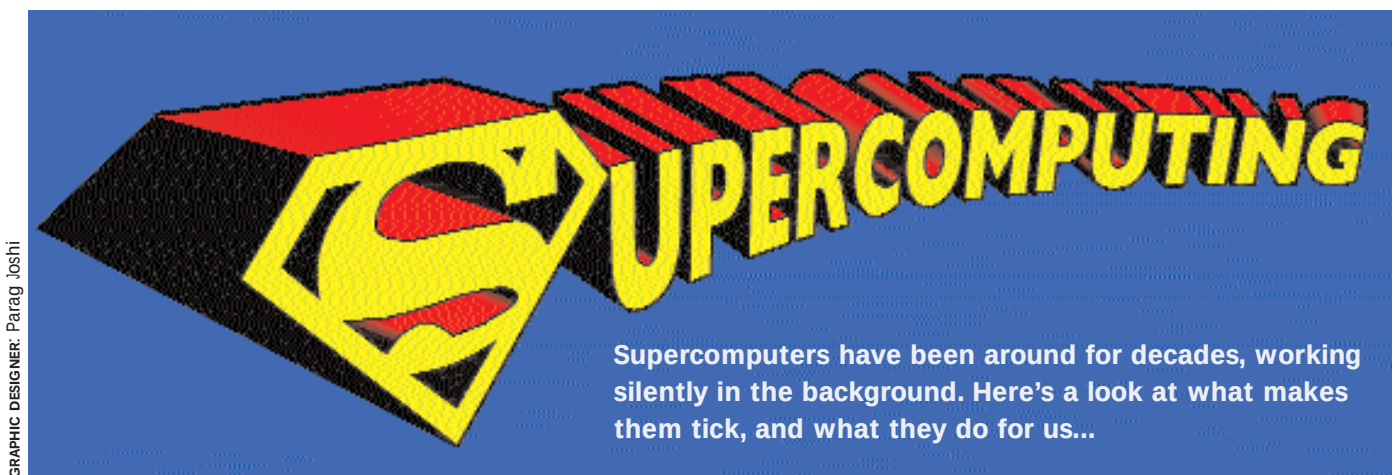
processors typically work with a few instructions within a clock cycle, using a technique known as pipelining.

Traditional supercomputers work on problems where several complex variables need to be analysed simultaneously; these problems may not be easy to break down into tiny, more easily analysable parts that could be worked upon in parallel. They involve large data sets and hence computers solving these problems need to have access to large shared memory spaces. The access itself has to be rapid, so that the computation process is not slowed down due to lack of data. Specialised processors and memory access technologies, with features to speed up specific types of calculations, solves that problem.

Another form of supercomputing is clustering. Clusters can be built by gang-ing up commonly used computers

(Computing) new Param Padma supercomputer is an example of a powerful supercomputing cluster, as is their older Param 10000.

Traditional HPC machines are specifically designed to increase performance, with optimised hardware and software. Based on the way instructions and data are manipulated, today's supercomputers can be classified as Multiple Input, Multiple Data (MIMD) machines. Single-processor MIMD machines can handle multiple streams of instructions and data streams. Multiple-processor MIMD machines are extremely efficient, as they run many sub-tasks in parallel. All this translates to a faster solution for the main task. Of course, MIMD is a broad classification and there are further sub-classifications depending on the memory organisation and access methodology—whether shared or distributed:



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Data Corporation), designing equipment for scientific computing. Cray's machines were the first (and the fastest) supercomputers of their time. They were designed to work on problems that were unsolvable, or computationally too tough for ordinary computers. Cray Inc, along with NEC, IBM, HP, SGI and others, make high performance supercomputers that have found new customers in industry-level research and development.

Not for your desktop

The architecture of a supercomputer is significantly different from that of a desktop machine. It is designed to handle large volumes of data, and to do tedious calculations. Supercomputing architectures have multiple processing nodes to handle parallel computing, whereas desktop

through a high-speed network. Of course, this is a rather simplistic way of defining these giants. Cluster-based HPC machines do not need to have access to a shared or common memory resource. Instead, the problems are broken up into tinier, parallelisable components. Within the cluster, each computer or node, has its own OS and memory. Each node works on a subsection of a problem that it can compute on its own—with minimal interaction of other nodes. This has become so popular that clusters are being adopted widely. However, computing clusters perform well only with obviously parallel problems, where data is easily broken up. There are still classes of problem that only traditional supercomputers can do. India's C-DAC's (Centre for Development of Advanced

Shared Memory MIMD (SM-MIMD): Within the shared memory approach, each of the multiple CPUs share the same memory address space. The entire memory can hence be accessed by all the CPUs, making it easier to program applications. However, there are also a couple of disadvantages to shared memory. One is memory contention, which occurs when two or more CPUs compete to access the same data and modify it. The problem grows in proportion with the number of processors, which is why there is a limit to the number of processors in SM-MIMD machines. Another problem is the technique by which the memory, as well as the CPUs are connected to each other. Ideally, the bandwidth of the memory should increase linearly, and processors should